

HW#5; Problems Statement:

- 4-1 Find the voltage gain v_O/v_S and current gain i_O/i_x in Figure P4-1 for $r = 5 \text{ k}\Omega$.

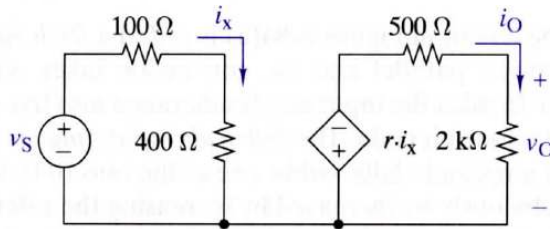


FIGURE P4-1

- 4-8 (a) Find an expression for the gain i_O/v_S in Figure P4-8 in terms of R_X .
(b) Select a value for R_X so that the gain is -0.227 .

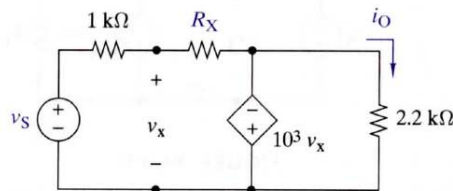


FIGURE P4-8

- 4-24 **D** Considering simplicity and standard 10% tolerance resistors as major constraints, design OP AMP circuits that produce the following voltage gains $\pm 10\%$: -100 , $+200$, $+1$, -0.5 , $+0.5$.
4-25 **E** Two OPAMP circuits are shown in Figure P4-25. Both claim to produce a gain of either ± 100 .
(a) Show that the claim is true.
(b) A practical source with a series resistor of $1 \text{ k}\Omega$ is connected to the input of each circuit. Does the original claim still hold? If it does not, explain why?

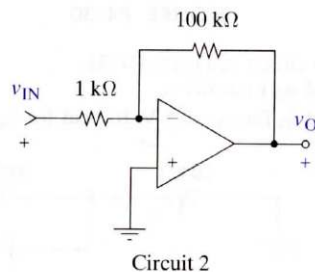
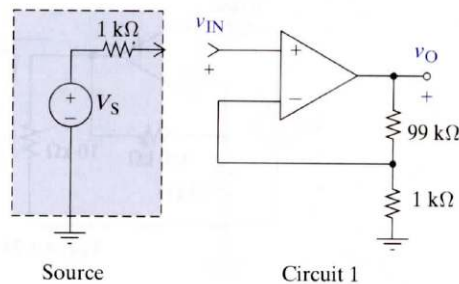


FIGURE P4-25

- 4-26 **E** Suppose the output of the practical source shown in Figure P4-25 needs to be amplified by -10^4 and you can use only the two circuits shown. How would you connect the circuits to achieve this? Explain why.

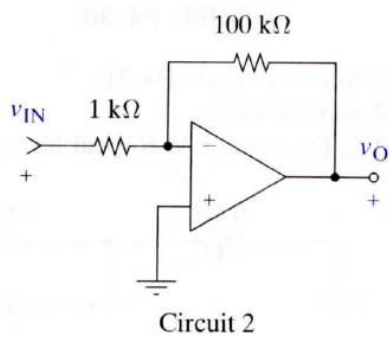
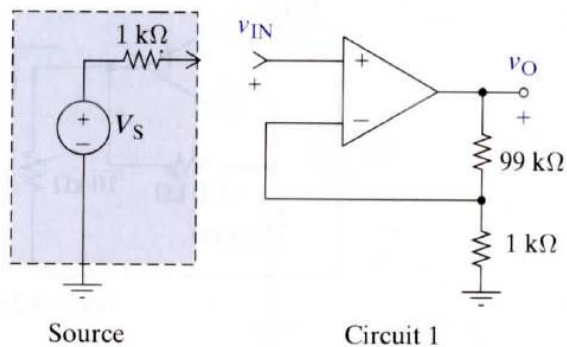


FIGURE P4-25

- 4-30 For the circuit in Figure P4-30:
- Find v_O in terms v_S .
 - Find i_O for $v_S = 1$ V. Repeat for $v_S = 3$ V.

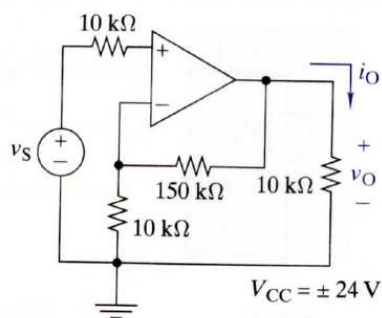


FIGURE P4-30

- 4-32 **E** A young designer needed to amplify a 2-V signal by the factors of 1, 5, and 10. Find the problem with the design shown in Figure P4-32. Recommend a fix.

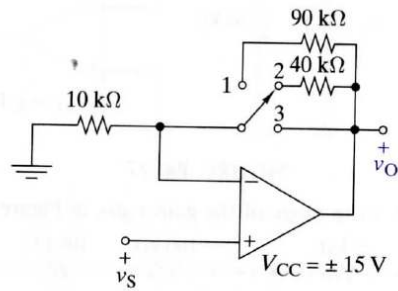


FIGURE P4-32

- 4-34 For the circuit in Figure P4-34:
- Find v_O in terms of the inputs v_1 and v_2 .
 - If $v_1 = 1$ V, what is the range of values v_2 can have without saturating the OP AMP?

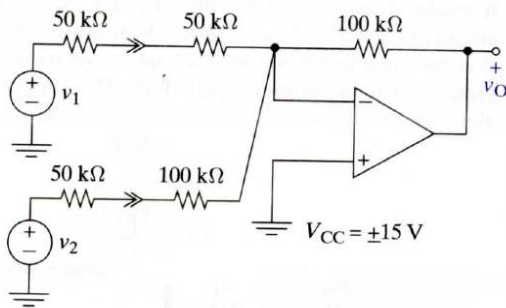


FIGURE P4-34

- 4-38 Find v_O in terms of v_{S1} and v_{S2} in Figure P4-38.

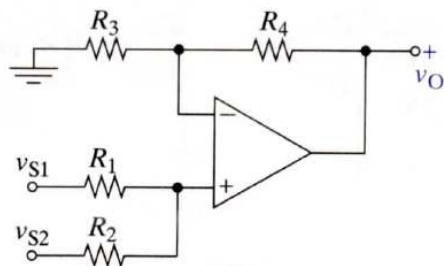


FIGURE P4-38

- 4-44 Find v_O in terms of the inputs v_{S1} and v_{S2} in Figure P4-44.

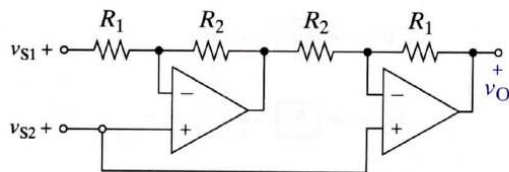


FIGURE P4-44